

APPENDIX

Simulations were performed using an agent-based model with timestep progression programmed in Python 3. A repository containing the code and results discussed in the paper is publicly available at <https://github.com/ENSwR>. This model simulates interactions between two key entities: a chemical species and a physiochemical space divided into localized environments (grid cells) (Figure A1). Different chemical species represent molecules with different biases for production by reactions in the local environment. The continuous process of chemical production is discretized into timesteps within the model. Furthermore, particles of a chemical species can chemically modify their local environment to promote the production of more particles of the same species. As such, local chemical environments possess properties conducive to different chemical species, and local environments can be modified when a chemical species is introduced in that environment. These modifications are implemented in the model through the impact of an individual on the value of “niche property” in its local environment. Periodically, the environment suffers a chemical ‘mass-extinction’ during a timestep. Such an event causes all species of chemical molecules to be removed from the whole environmental space (i.e., flushed or destroyed). However, the localized environments are not physically destroyed in this process, allowing prior chemical modifications of the physical environment to persist until the chemical process of molecule production is re-established in each microenvironment.

This model has independent settings to enable or disable resource limitation, particle decay, and migration conditions. Using combinations of these conditions, we can create eight (2^3) unique modes of simulation.

Chemical species production under unlimited resources

When resources are unlimited, creation of the individual particles of a chemical species is determined by the sum of its species-specific production bias and the chemical properties of the environment (Table A1, Equation 1). In this scenario, production of one chemical species does not affect the production of another chemical species, and there are no imposed maximum limits on particle production rate for any of the chemical species.

$$\# \text{ of } X \text{ Particles Formed} = X_{\text{production_bias}} + \text{Environment}_{X_niche_property} \quad (1)$$

Chemical species production under limited resources

In this scenario, the production of a chemical species is limited by the availability of shared precursor molecules. These are handled generically within the model by a generalized “resources” parameter. Resource utilization is modeled independently within each localized environment. In our simulations, each localized environment replenishes to a fixed number of resource “doses” every timestep, which reflects the baseline chemical production of precursor molecules within the local environment. Each dose of resource is chemically transformed to produce one chemical species of a specified type, such that each localized environment will spontaneously produce a total of five chemical particles during each non-extinction timestep of the model. This process is modeled as a Bernoulli experiment occurring each timestep. In these experiments, consumption of

each resource “dose” is treated as a trial where probability of outcomes can be determined by the accumulated effects of a chemical species’ relative rate of baseline production (bias) and the niche-constructed chemical properties of its environment (Equation 2). This imposes resource competition for production of different chemical species.

For each Bernoulli trial at grid cell n, m (2)

$$P(A)_{n,m} = \frac{A_{production_bias} + A_{niche_property_{n,m}}}{(A_{production_bias} + A_{niche_property_{n,m}}) + (B_{production_bias} + B_{niche_property_{n,m}})}$$

$$P(B)_{n,m} = \frac{B_{production_bias} + B_{niche_property_{n,m}}}{(A_{production_bias} + A_{niche_property_{n,m}}) + (B_{production_bias} + B_{niche_property_{n,m}})}$$

$$P(A)_{n,m} + P(B)_{n,m} = 1$$

Niche Construction

Niches in this simulation are assumed to be unique to their supported chemical species without overlaps or exclusions. While implementation of niche construction is agnostic to resources and potential competition, the niche impact of an individual niche constructor diminishes proportionally to the environment’s existing niche-property value (Equation 3). This prevents compounding exponential growth and asserts the assumption that “well-developed” niches become harder to improve.

(3)

$$Individual\ impact_t = \begin{cases} X_{niche_construction} & , \quad X_{niche_property_{n,m,t}} = 0 \\ \frac{X_{niche_construction}}{X_{niche_property_{n,m,t}}} & , \quad X_{niche_property_{n,m,t}} > 0 \end{cases}$$

Particle Decay

Decay reflects the capacity by some molecules to have greater molecular stability than others and therefore to persist longer. In these models, chemical species have a lifespan parameter which specifies how many timesteps a particle of that type can persist. When a particle has exceeded its lifespan, it is considered to have decayed, and it is removed from the environment. Without decay, chemical particles persist until they are destroyed in a mass-extinction timestep.

Diffusion (an analogue to “migration”)

Chemical diffusion is well understood, and because it is mathematically convenient, it is often used as a model to approximate large scale patterns of animal movement. Here we are looking at something else, i.e., movement of particles caused by a mechanical force in the environment. We call this “particle diffusion” simply to refer to the partially random movement of chemical species. We implement diffusion via a random process of individual particle movement between neighbouring local environments (grid cells). In this model, particle movement is a discrete process coordinated by the SimEngine at discrete time points (*i.e.*, we discretized and marginalized the

effect of the continuous process). We update particle movement between local environments during each mass-extinction timestep. At this time point, each particle has a 1% chance of surviving the extinction event, but gets relocated to an adjacent localized environment. These new arrivals immediately interact with their new local environment to modify the relevant chemical niche parameter(s). As described above, any modifications made to a local environment's properties persists within that local environment.

Additionally, for the purposes of demonstrating dynamics in an extremely challenging case, particle species "A" was assigned a positive production bias only within a single local environment. Hence, for species "A" to populate new local environments, individual A's must "diffuse" to apply their species' unique modification to the local chemical niche-property. As its production bias value is 0 in any new environment, all future production of "A" in a new environment is determined by the effect of its niche construction (implemented as the value of its "niche" parameter). Effectively, without "diffusion" enabled, "A" is restricted to a single local environment within the grid.

Running the Model

The model is specified according to user-defined settings in a configuration file. Running a simulation according to the specified model requires calling the *run_sim* module with the pathway for the configuration file as its argument. Parameters in the configuration file are described in Table A1. The *run_sim* module uses the configuration file to construct a model object. The model is then sent to the *sim* module, which creates the environmental grid and begins computing the simulation timesteps. In doing so, the *sim* module (1) invokes a *SimEngine* module responsible for probability calculations, (2) coordinates the environment and particle objects to perform the steps required for the simulation, and (3) logs the data to be sent back to the *sim* module, where it is formatted and written to files after all timesteps are completed (Figure A1).

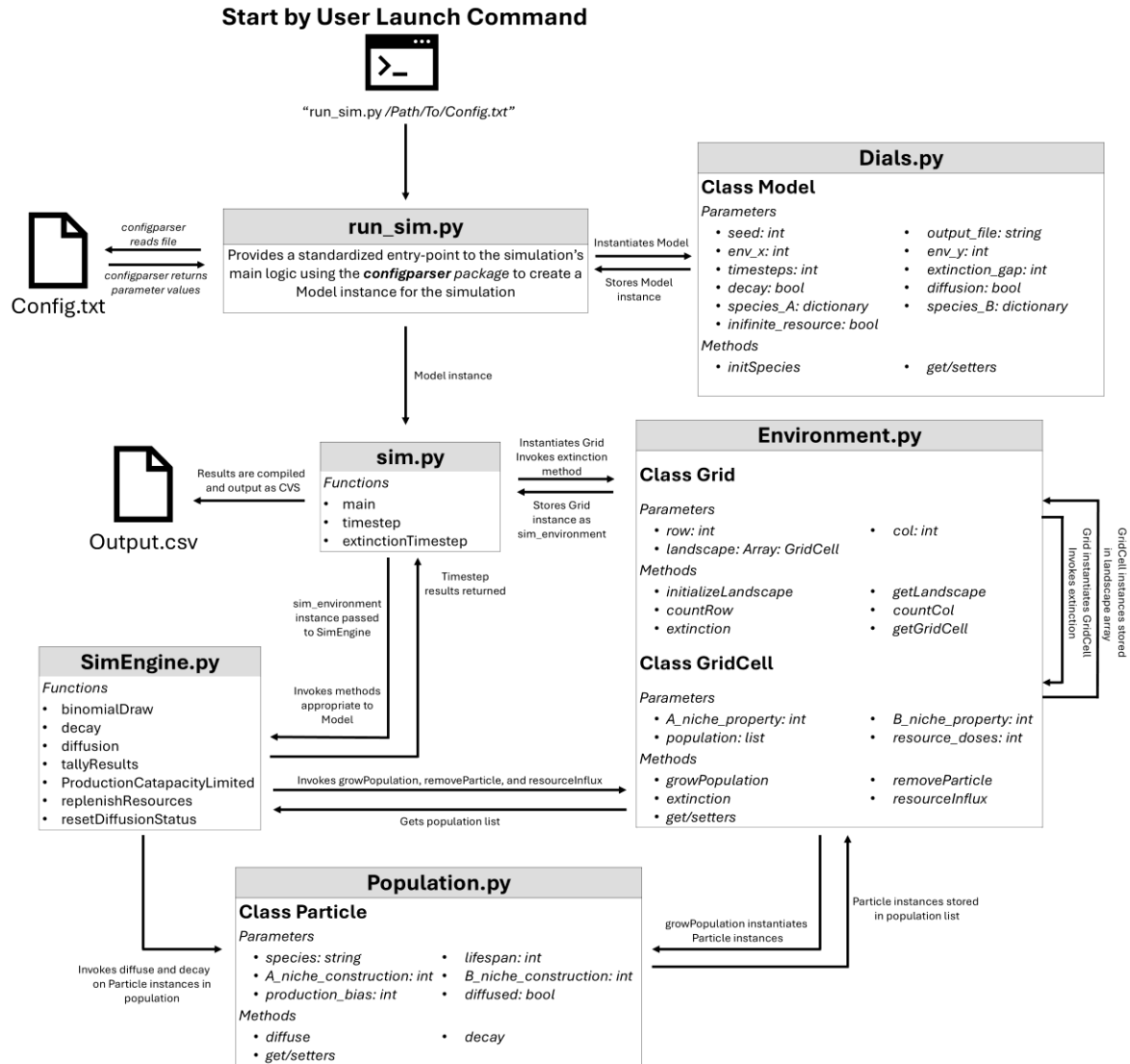


Figure A1 – Relationship diagram depicting flow of information through interactions between the modules and classes of the ENSwR simulation to produce output.

Table A1 – Description of all parameters comprising a configuration file.

Name	Data Type	Description
<i>[Dials]</i>		Header for model settings
seed	int	Optional declaration for random number seed
timesteps	int	Number of timesteps to run the simulation
extinction_gap	int	Timesteps between mass-extinctions
environment_x	int	Size of Environment grid (x dimension)
environment_y	int	Size of Environment grid (y dimension)
output_file	string	Filepath to direct output file
decay	bool	Toggle whether model includes Particle decay
diffusion	bool	Toggle whether to include diffusion
infinite_resources	bool	Toggle whether to include resource limitation
<i>[Type_A/B]</i>		Header for Particle types (A or B) settings
species	string	Species of Particle (A or B)
A/B_niche_construction	int	Niche construction value for species
production_bias	int	Innate production for species
lifespan	int	Timesteps to persist in decay models